PHONE CLUSTER AI LAB - CHECKPOINT 2026-11-06
Overall readiness: 55% [############---------]

WHAT WE FOUND
- The video shows distributed LLM inference, not a general-purpose wireless GPU.
- Best first open-source route: Pooled (WebGPU + WebRTC in browser tabs).
- Current Pooled docs show iPhone Safari can join and hold model layers.
- Devices without WebGPU can remain ask-only/controller devices.
- Local API bridge: OpenAI/Anthropic compatible on 127.0.0.1:8080.
- Starter baseline: Qwen3 1.7B; about 4 GB room memory at 8K context,
  about 5.6 GB at 16K context.

ALREADY BUILT
- GitHub project structure and research notes.
- Windows preflight, Pooled bootstrap, API serve, and verification scripts.
- Pinned Pooled upstream baseline for reproducibility.
- Worker vs Controller mobile design.
- Z Cluster mobile compatibility/PWA shell.
- Progress tracker and hardware benchmark template.

MOBILE EXPERIENCE
GPU Worker Candidate: browser exposes WebGPU and returns a usable GPU adapter.
Controller: WebGPU unavailable; device still handles chat/control workflows.

FIRST REAL TEST
PC + one iPhone/iPad -> same Pooled room -> Qwen3 1.7B -> ask:
Reply with exactly: PHONE-CLUSTER-OK
Do not mark hardware tested until this passes on the actual devices.

BUILD METHOD AND RESUME POINT

KARPATHY / EVIDENCE-FIRST
- Start with the smallest measurable baseline.
- Change one variable at a time.
- Verify every change and preserve a rollback path.
- Never replace a real test with an ’it should work’ claim.

CAVEMAN MODE
- One objective per step.
- Simple language, obvious buttons/commands, visible PASS/FAIL.
- When troubleshooting, reduce the system until the failure is isolated.

ENGAGEMENT
- Always show device role, model, download state, room/API status, and next action.
- Avoid dead screens and unexplained waiting.

RESUME - DO NOT RESTART RESEARCH
1. Run PC + one iPhone/iPad baseline and get PHONE-CLUSTER-OK.
2. Run local API bridge and verify chat/completions.
3. Connect AnythingLLM or a coding agent.
4. Add a second mobile device and benchmark.
5. Test a larger model only after baseline stability.
6. Expand the Z Cluster dashboard.

GITHUB
ZeusGriffin/open-design/projects/hardware/phone-cluster-ai

SOURCES
https://github.com/Nehanth/pooled
https://pooled.run
https://www.youtube.com/watch?v=reOeczuEzOA